When the Dog Has Inertia: Feedback Equilibria, Lost Potential and Equilibrium Selection in an Influence Game

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Abstract

Two players repeatedly pull a shared point, the dog, toward their own houses. Without inertia the game is an instance of the Battling Influencers Game, an exact potential game (Wu et al., 2025). We study what survives when the dog keeps part of its position each round. We prove that holding the one-shot equilibrium actions is a Markov-perfect equilibrium if and only if each player's slack beyond the dog is at least w0(1 - g w0)/(1 - g w0^2), where w0 is the inertia and g the discount, and we give an explicit piecewise-affine feedback equilibrium whenever the houses lie on either side of the dog; numerically, a fixed point for the players' targets gives a feedback equilibrium in 199 of 200 random games. The condition extends to many houses, where the admissible inertia shrinks like 1/N. We show that the dynamic game is not a Markov potential game once policies react to the dog. For learning, we find that policy-gradient self-play from random starts learns to hold the one-shot actions, which is exploitable off the equilibrium path, and that on the potential's flat direction learning is a random walk stopped by the action box, so which of two mirror equilibria wins is a fair coin. Exact gradient play finds the feedback equilibrium with a table of actions but not with a shared network.

1. Introduction

Influence games, in which agents with fixed targets push a shared quantity, are a clean testbed for multi-agent learning: the static game has a potential, its equilibria are computable, and its learning dynamics can be checked exactly. The Battling Influencers Game proves the potential structure and characterizes pure equilibria of the one-shot game (Wu et al., 2025). Real influence is persistent, though: opinions, prices or a dog on a leash keep part of their state. This paper asks what the static results imply once the shared point has inertia, and what policy-gradient learners find. Our contributions, each verified computationally in the accompanying repository:

- A necessary and sufficient condition for the one-shot equilibrium to remain Markov-perfect under inertia (Proposition 1), and an explicit piecewise-affine feedback equilibrium when it does not (Proposition 2), both with proofs and checked against an exact dynamic program.
- The extension of the condition to many houses, where the static outcome is the generalized median of the average voting rule (Renault and Trannoy, 2005), and the finding that the admissible inertia shrinks like 1/N.
- A concrete check that the dynamic game is not a Markov potential game, although every state game is a potential game.
- Equilibria in target form beyond Proposition 2, found by a two-dimensional fixed point, and the continuous-time limit, where the equilibrium is bang-bang and a noisy dog moves its switching points.
- Two learning findings: self-play misses the feedback equilibrium, and selection along a flat potential is a random walk stopped by the action box. The second corrects an earlier small-sample result of ours.

The static potential structure and the general fact that stage-wise potential games need not be Markov potential games are known; we claim novelty only for the results above, for this game, as far as the literature

we checked goes.

2. Related work

Potential games go back to Monderer and Shapley (1996); noisy learning in them selects potential maximizers (Blume, 1993). The Battling Influencers Game (Wu et al., 2025) is the static game studied here. Markov potential games (Leonardos et al., ICLR 2022) give convergence of independent policy gradient, and a Markov game whose stage games are potential games need not be one; a continuous-state counterexample to sufficient conditions appears in arXiv 2405.08206. Potential differential games reduce to one optimal-control problem in open loop (Kavuncu, Yaraneri and Mehr, RSS 2021). Feedback equilibria of constrained linear-quadratic games are piecewise affine over polyhedral regions (arXiv 2001.01826; Benenati and Belgioioso, arXiv 2512.07749); Proposition 2 is an explicit instance with a proof specific to this game. The average voting rule's equilibrium outcome is a generalized median (Renault and Trannoy, 2005). Policy gradient provably finds Nash equilibria of unconstrained linear-quadratic games (Hambly, Xu and Yang, JMLR 2023); equilibrium selection by multi-agent policy gradient is studied in finite games (ICML 2026, arXiv 2605.18078).

3. Model

Two players choose a_red and a_blue in the unit square. The dog moves to x' = w0 x + (1 - w0)(w a_red + (1 - w) a_blue), with inertia w0 in [0, 1) and red's weight w in (0, 1), and player i receives -||x' - h_i||^2, discounted by g in [0, 1). Rewards, dynamics and the action box all separate by coordinate, so the game splits into two independent one-dimensional games, and a profile is a feedback (Markov-perfect) equilibrium iff it is one in each coordinate. With w0 = 0 this is the one-shot game, with potential Phi = -||d||^2 + 2 w h_red . a_red + 2 (1 - w) h_blue . a_blue, where d = w a_red + (1 - w) a_blue. Let (a_red*, a_blue*) be the one-shot equilibrium, d* its dog, k = (1 - g w0^2)/(1 - g w0) and T_i = d* + k (h_i - d*) player i's target.

4. Feedback equilibria with inertia

Proposition 1. Holding the one-shot actions at every dog position is a feedback equilibrium iff every player at an edge of the box has slack s >= rho(w0, g) = w0 (1 - g w0)/(1 - g w0^2), with s = (h - d*)/(1 - d*) at the top edge and (d* - h)/d* at the bottom, and, when w0 > 0, no one-shot action is strictly inside the box.

Proposition 2. If the houses lie on either side of the dog (h_blue <= w <= h_red after relabeling), the policy in which each player aims the next dog position at its target T_i, as far as its own weight allows, with the other at its one-shot action, is a feedback equilibrium for every w0 in [0, 1) and g in [0, 1). It is piecewise affine in the dog's position and equals the one-shot profile wherever holding is optimal.

Both proofs use the one-shot deviation principle. Under holding, a player's one-round objective is a concave quadratic in the next dog position peaked at T_i, which gives Proposition 1. For Proposition 2, a contraction argument shows the one-round objective under the policy is single-peaked at T_i on all of [0, 1], so the clipped target is a best reply. Full proofs are in docs/INERTIA_PROOFS.md.

Checks. An exact dynamic program (Nash value iteration, 401 positions by 101 actions per coordinate) agrees with both propositions to one action-grid step in every coordinate where the houses differ, in five house setups at w0 = 0.3, 0.6 and 0.9 (where the houses coincide, the game has a continuum of equilibria and the program picks another). On 200 random one-coordinate games, with the deviation gain computed by simulating the dog rather than on a grid:

One-shot equilibrium	Games	Explicit policy is an equilibrium (gain < 1e-9)	Largest one-round gain	Holding is Markov-perfect
straddle	77	77 of 77	0.0e+00	30 of 77
interior	123	70 of 123	2.3e-02	0 of 123

Proposition 2 holds in all 77 straddling games, and holding alone in 30 of them, as Proposition 1 predicts. With an interior one-shot action the policy is an equilibrium in only 70 of 123 games: the interior player pushes at an edge off its house, the dog returns faster than under holding, and the other player's best target moves. Because a player's one-round objective depends only on the next dog position, each player can still aim at one target everywhere when that objective is single-peaked; with the targets from a two-dimensional fixed point:

One-shot equilibrium	Games	Proposition 2's policy	Target fixed point	Fixed point: largest gain	Fixed point: iterations (max)
straddle	77	77 of 77	77 of 77	0.0e+00	33
interior	123	70 of 123	122 of 123	1.4e-05	32

The fixed point is a feedback equilibrium in 199 of 200 random games; a proof, and uniqueness, are open.

Many houses. With N players the one-shot dog is the generalized median of the average voting rule, at most one player (the pivot) is inside the box, and Proposition 1 holds verbatim. Drawing N houses from Beta(2, 5) with equal weights (200 draws per N, discount 0.9):

Houses N	One-shot dog: mean (sd); limit 0.339	No pivot inside (limit 0.34)	Largest inertia: mean	N times it, median when no pivot
2	0.368 (0.115)	0.20	0.0399	0.27
4	0.340 (0.090)	0.28	0.0357	0.43
8	0.340 (0.055)	0.34	0.0206	0.27
16	0.337 (0.038)	0.29	0.0111	0.45
32	0.340 (0.028)	0.35	0.0058	0.33
64	0.341 (0.019)	0.32	0.0024	0.32
128	0.339 (0.013)	0.34	0.0012	0.34
256	0.340 (0.010)	0.34	0.0007	0.33
512	0.340 (0.007)	0.28	0.0003	0.43
1024	0.340 (0.005)	0.32	0.0002	0.35

A pivot inside the box (about two draws in three) rules holding out at any inertia; otherwise the largest admissible inertia scales like 0.35/N. With many houses the static equilibrium is almost never a feedback equilibrium. The target form carries over: if every player aims at a target, each round is the one-shot game with the targets as houses, and fixed-point targets give a feedback equilibrium whenever every player's one-round objective is single-peaked:

Players	Games	Target form is an equilibrium	with single-peaked objectives	with several peaks	Largest gain	Holding is Markov-perfect
3	20	20 of 20	19 of 19	1 of 1	4.8e-07	4 of 20
5	20	18 of 20	14 of 14	4 of 6	2.6e-04	0 of 20
8	20	17 of 20	9 of 9	8 of 11	5.7e-04	1 of 20

5. The dynamic game is not a Markov potential game

In a Markov potential game, every 4-cycle of unilateral policy changes sums to zero. Exact values from the starting dog position of one setup:

Inertia w0	Policies that ignore the dog	Policies that react to the dog
0.0	+6.7e-15	+7.90
0.3	+3.1e-15	+7.12
0.6	+1.8e-15	+5.63
0.9	+5.6e-15	+1.85

With policies that ignore the dog, the dynamic game remains a potential game at any inertia (open loop); with policies that react to it, it is not, even at w0 = 0. The convergence theory of independent policy gradient in Markov potential games therefore does not cover the feedback play studied below.

6. Learning

Self-play misses the feedback equilibrium. We train self-play from uniformly random dog positions and score the learned mean policy by its feedback exploitability: the largest and mean gain, over dog positions and players, from a one-round deviation followed by the learned policy, with the dog simulated exactly. Every feedback equilibrium scores 0, so the score does not assume uniqueness.

Setting	Policy	Runs	Largest one-round gain: median (range)	Mean one-round gain: median
setup 3, w0 = 0.6	Proposition 2	1	0.0000	0.0000
setup 3, w0 = 0.6	hold the one-shot actions	1	0.0756	0.0114
setup 3, w0 = 0.6	REINFORCE	5	0.1537 (0.1156 to 0.6714)	0.0275
setup 3, w0 = 0.6	A2C	5	0.0756 (0.0755 to 0.0759)	0.0115
setup 3, w0 = 0.6	PPO	5	0.0756 (0.0756 to 0.4599)	0.0115
setup 3, w0 = 0.6	REINFORCE, Beta policy	5	0.0900 (0.0855 to 0.1287)	0.0247
setup 3, w0 = 0.6	A2C, Beta policy	5	0.0854 (0.0813 to 0.0891)	0.0222
setup 3, w0 = 0.6	PPO, Beta policy	5	0.0991 (0.0700 to 0.4077)	0.0323
setup 3, w0 = 0.6	A2C, 15000 episodes	3	0.0756 (0.0756 to 0.0756)	0.0114
setup 3, w0 = 0.6	A2C, Beta policy, 15000 episodes	3	0.0791 (0.0776 to 0.0798)	0.0150
setup 5, w0 = 0.9	Proposition 2	1	0.0000	0.0000
setup 5, w0 = 0.9	hold the one-shot actions	1	0.2084	0.0397
setup 5, w0 = 0.9	REINFORCE	5	0.2014 (0.1506 to 0.3832)	0.0337
setup 5, w0 = 0.9	A2C	5	0.2084 (0.2084 to 0.2084)	0.0397
setup 5, w0 = 0.9	PPO	5	0.2084 (0.2084 to 0.2084)	0.0397

A2C and PPO learn to hold the one-shot actions at every position, which is unexploitable on the equilibrium path and exploitable off it, and this survives five times longer training, a non-saturating Beta policy and a setting with 3.5 times the off-path incentive. REINFORCE does not do better. To separate the learning dynamics from estimation noise, we also run exact gradient play on one coordinate: deterministic policies, each player ascending the exact gradient of its own payoff averaged over starting positions.

Setting	Policy	Runs	Within 0.05 of Proposition 2	Within 0.05 of holding	Largest one-round gain: median (range)
setup 3 x, w0 = 0.6	table, clamp	1	1 of 1	0 of 1	0.0000
setup 3 x, w0 = 0.6	table, sigmoid	1	1 of 1	0 of 1	0.0007
setup 3 x, w0 = 0.6	network, 16 units	5	0 of 5	1 of 5	0.0479 (0.0479 to 0.0756)
setup 3 x, w0 = 0.6	network, 64 units	5	0 of 5	0 of 5	0.0479 (0.0479 to 0.0479)
setup 5 x, w0 = 0.9	table, clamp	1	1 of 1	0 of 1	0.0000
setup 5 x, w0 = 0.9	table, sigmoid	1	1 of 1	0 of 1	0.0000
setup 5 x, w0 = 0.9	network, 16 units	5	0 of 5	2 of 5	0.1042 (0.1042 to 0.1042)
setup 5 x, w0 = 0.9	network, 64 units	5	0 of 5	1 of 5	0.1042 (0.1042 to 0.1042)

A table of actions reaches the feedback equilibrium every time and a shared network never does, so function approximation, not sampling noise, is what stops the learners; a plausible but untested mechanism is that the network carries "push to the edge" from the many positions where it is right to the few where easing off is right. Low on-path exploitability therefore does not certify a feedback equilibrium.

Selection on a flat potential. When the houses share a coordinate, the potential is flat along a continuum of equilibria. REINFORCE self-play, 50 seeds per initial exploration noise:

Initial noise (std)	red top, blue bottom	red bottom, blue top	both at y = 0.5	other	p (mirror images equally likely)
0.14	19 of 50	22 of 50	0 of 50	9 of 50	0.76
0.37	31 of 50	19 of 50	0 of 50	0 of 50	0.12
0.61 (default)	31 of 50	19 of 50	0 of 50	0 of 50	0.12
1.00	30 of 50	20 of 50	0 of 50	0 of 50	0.20

No run ends at the interior equilibrium, and which mirror image wins is consistent with a fair coin; an earlier 10-seed analysis of ours had suggested that the noise level flips it, which did not replicate. Two ablations show why:

Policy mean in y, actions and dog	red top, blue bottom	red bottom, blue top	both at y = 0.5	other	Red's final y: quartiles	Red's y outside [0, 1]
sigmoid, clipped (default)	31 of 50	19 of 50	0 of 50	0 of 50	0.00 / 0.99 / 1.00	0 of 50
sigmoid, not clipped	20 of 50	21 of 50	0 of 50	9 of 50	0.03 / 0.64 / 0.97	0 of 50
linear, not clipped	17 of 50	20 of 50	0 of 50	13 of 50	-2.26 / 0.33 / 1.29	38 of 50

The expected gradient along the flat direction is zero but noisy, so the policy means random-walk; with an unbounded mean the walk leaves the box, and the default sigmoid policy stops it at the edges. Clipped exploration noise also makes the averaged potential dip at the interior, pushing either way equally.

7. Continuous time and noise

As the time step dt shrinks (w0 = 1 - dt, discount exp(-r dt)), k tends to (r + 2)/(r + 1) and rho to (r + 1)/(r + 2): holding the one-shot actions needs slack at least (1 + r)/(2 + r), and Proposition 2's equilibrium becomes bang-bang, each player pushing its edge toward its target and switching there. With a Gaussian shock on the dog the equilibrium stays bang-bang, but its switching points move with the noise, also without clipping, which a linear-quadratic benchmark with an action cost would not show (its policies are certainty-equivalent):

Dog	Time step	Noise sigma	Red switches at	Blue switches at	Value iteration
closed form (Proposition 2)	0.05	0	0.718	0.520	-
closed form (Proposition 2)	0.025	0	0.709	0.527	-
continuous-time limit	0	0	0.700	0.533	-
clipped to [0, 1]	0.05	0.0	0.725	0.520	554 sweeps
clipped to [0, 1]	0.05	0.1	0.760	0.530	550 sweeps
clipped to [0, 1]	0.05	0.2	0.765	0.545	559 sweeps
clipped to [0, 1]	0.05	0.4	0.790	0.585	did not converge
clipped to [0, 1]	0.025	0.0	0.710	0.530	1107 sweeps
clipped to [0, 1]	0.025	0.2	0.765	0.550	1117 sweeps
unclipped	0.05	0.0	0.725	0.520	554 sweeps
unclipped	0.05	0.02	0.730	0.525	551 sweeps
unclipped	0.05	0.05	0.740	0.525	548 sweeps
unclipped	0.05	0.1	0.760	0.530	550 sweeps
unclipped	0.05	0.2	0.765	0.545	559 sweeps
unclipped	0.05	0.4	0.775	0.585	605 sweeps

The shift grows smoothly from zero noise and is the same with half the time step and with an unclipped dog; why noise moves the switching points this way is open, and one clipped run at sigma = 0.4 did not converge.

8. Discussion and limits

The static potential structure does not carry over to feedback play, but the static equilibrium survives on the equilibrium path, and Proposition 1 says exactly where it fails off the path. The feedback equilibrium is explicit when the houses straddle the dog and has the same target form numerically when they do not; learners nevertheless converge to the static profile. Limits: the proofs are ours and not refereed; Proposition 2 excludes interior one-shot actions, where the target fixed point is only numerical; learning results use one game, a few settings and five seeds per setting; the continuous-time statements are limits of discrete-time equilibria, and continuous-time q-learning is not studied.

References

Blume (1993), The statistical mechanics of strategic interaction, Games and Economic Behavior. Benenati and Belgioioso (2025), arXiv 2512.07749. Fudenberg and Tirole (1991), Game Theory. Hambly, Xu and Yang (2023), Policy gradient methods find the Nash equilibrium in N-player general-sum linear-quadratic games, JMLR. Kavuncu, Yaraneri and Mehr (2021), Potential iLQR, RSS. Leonardos, Overman, Panageas and Piliouras (2022), Global convergence of multi-agent policy gradient in Markov potential games, ICLR. Monderer and Shapley (1996), Potential games, Games and Economic Behavior. Renault and Trannoy (2005), Protecting minorities through the average voting rule, Journal of Public Economic Theory 7, 169-199. Wu, Zhu, Cai and Zhu (2025), Battling Influencers Game, arXiv 2502.01127. arXiv 2001.01826 (feedback Nash equilibria of constrained dynamic games). arXiv 2405.08206 (Beyond Theorems: a counterexample to potential Markov game criteria). arXiv 2605.18078 (equilibrium selection in multi-agent policy gradients via opponent-aware basin entry, ICML 2026).